

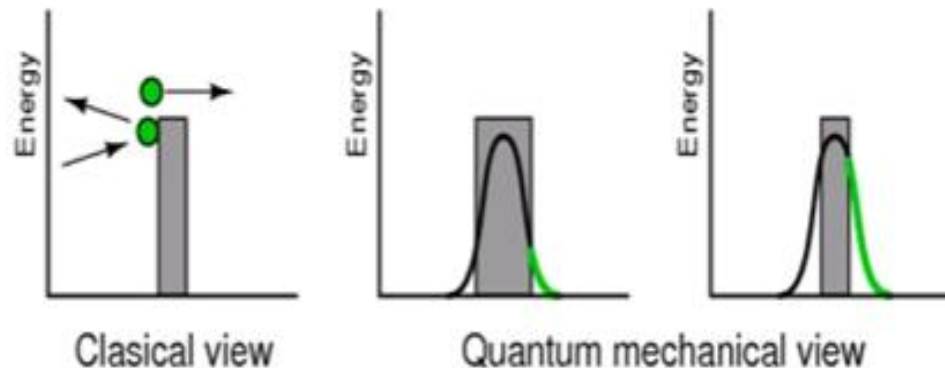
Unit-4
Lecture-4

**Application of Quantum tunnelling effect
on
Scanning Tunnelling Microscope.**

Scanning Tunnelling Microscope (STM)

Introduction

- A **scanning tunneling microscope (STM)** is an instrument for imaging surfaces at the atomic level.
- STM is based on the concept of quantum tunneling.
- In STM good resolution is considered to be 0.1 nm lateral resolution and 0.01 nm (10 pm) depth resolution.
- Quantum mechanics explain the behaviour of photons and electrons.
- Quantum Tunneling means tunneling current occurs when electrons move through a barrier that they classically shouldn't be able to move through. In classical terms, if you don't have enough energy to move "over" a barrier, you won't.



Quantum Tunneling

Classical Mechanics



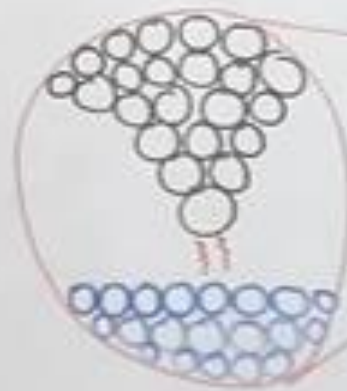
Small particles
Like Electrons

shows wave character
it easily penetrate or move
from one point to another

Quantum Mechanics

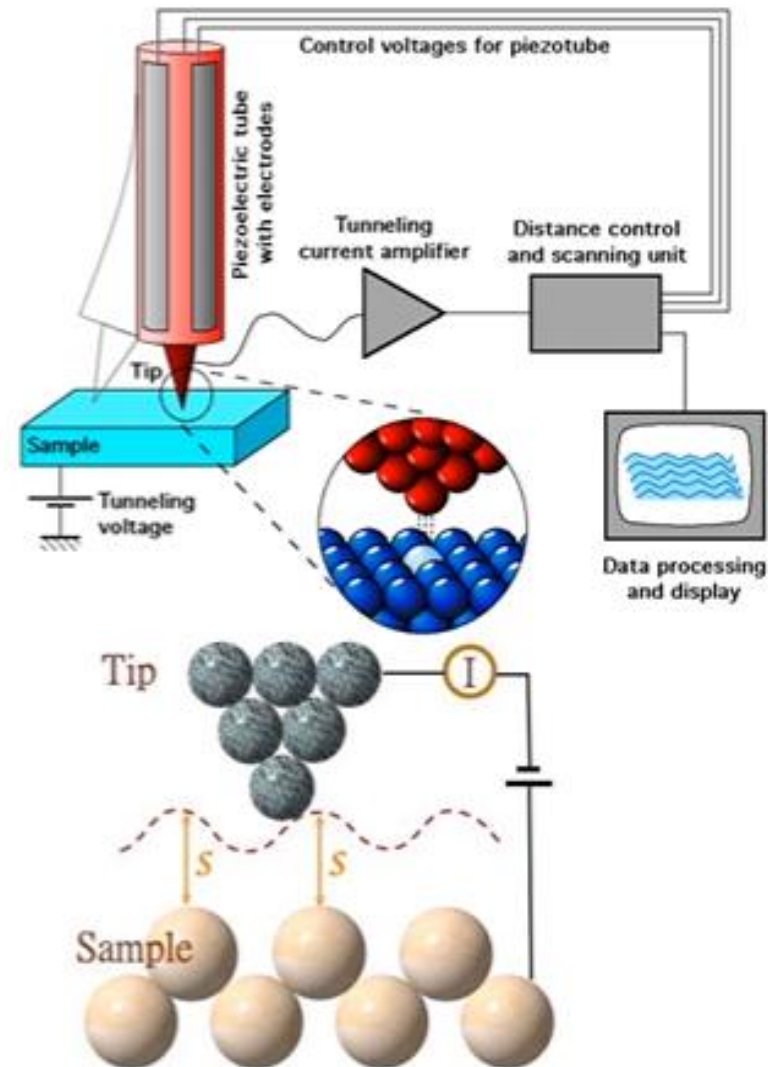


Like an
Electron



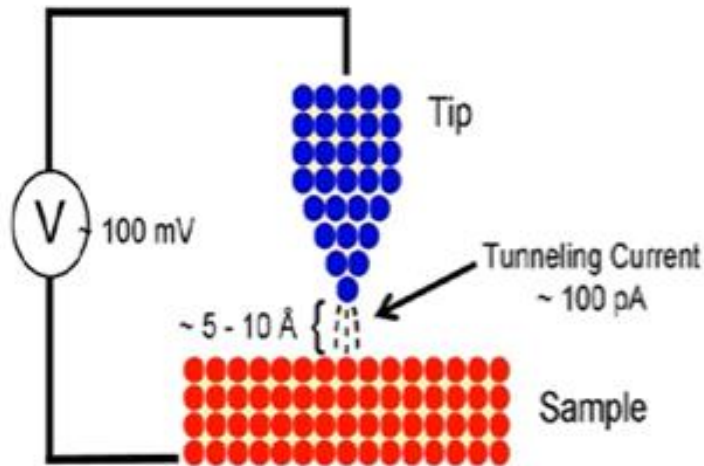
Construction

- ✓ Piezoelectric tube with electrode
- ✓ Tunneling Voltage
- ✓ Scanning Tip
- ✓ Tunneling Current Amplifier
- ✓ Distance control and scanning unit
- ✓ Data processing and Display
- ✓ Sample

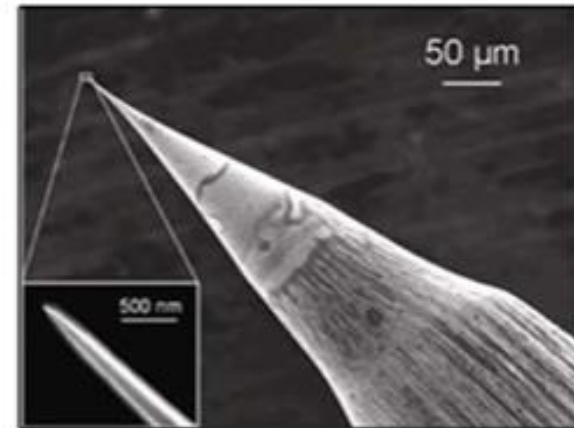


1. Scanning Tip

- STM tips are usually made from tungsten metal or a platinum-iridium alloy where at the very end of the tip (called apex) there is one atom of the material.
- Scanning tip is the most important aspect of the STM as tunneling current is carried by that particular atom



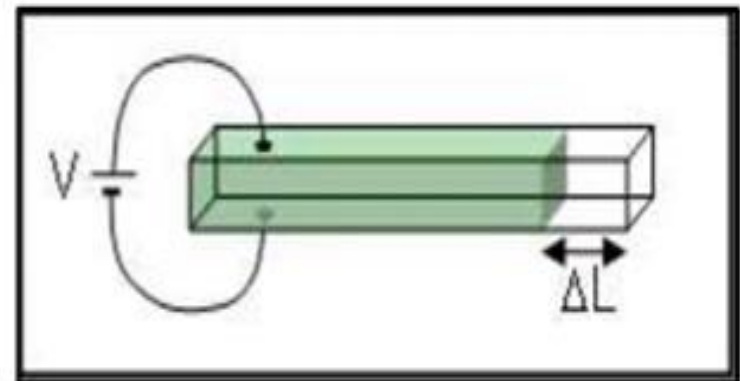
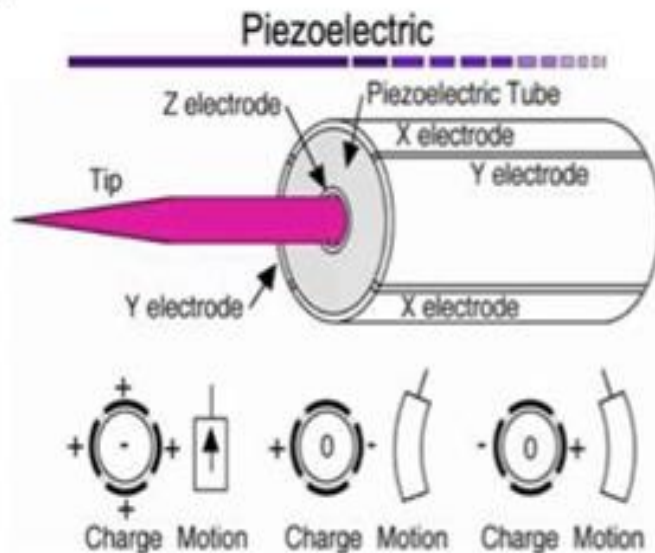
Sample and Tip at atomic level



STM tip under $10^3\times$ and $10^5\times$ magnification

2. Piezoelectric Scanner

- The scanner tip is attached to a piezoelectric tube scanner.
- Piezoelectric the effect is a phenomenon under which the material changes its length accordingly when put under an electrical voltage.
- By adjusting the voltage on the piezoelectric element, the distance between the tip and the surface can be regulated.
- Piezoelectric crystals expand and contract very slightly depending on the voltage applied to them and this principle is used to control the horizontal position x , y , and the height z of the scanning tip.



3. Distance Control And Scanning Unit

- Position control using piezoelectric means is extremely fine, so a coarse control is needed to position the tip close enough to the sample before the piezoelectric control can take over.

4. Data Processing Unit(Computer)

- The computer records the tunneling current and controls the voltage to the piezoelectric tubes to produce a 3-dimensional map of the sample surface

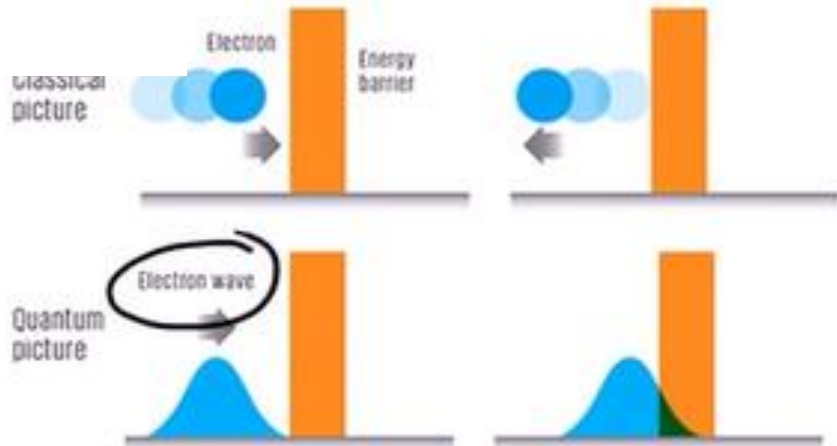
5. Vibration Isolation System

- STM deals with extremely fine position measurements so the isolation of any vibrations is very important.
- The tip and surface distance must be maintained in $\text{\AA}$ (0.1 nm) to get desired atomic resolution.
- Due to extremely high sensitivity of tunneling current between tip and sample surface height, it is absolutely necessary to reduce inner vibrations and to isolate the system from external vibration.
- Damping can be achieved by :
 - ❖ Pneumatic systems
 - ❖ Spring system
 - ❖ Eddy current system

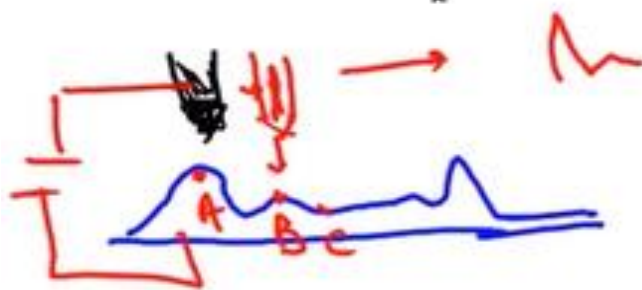
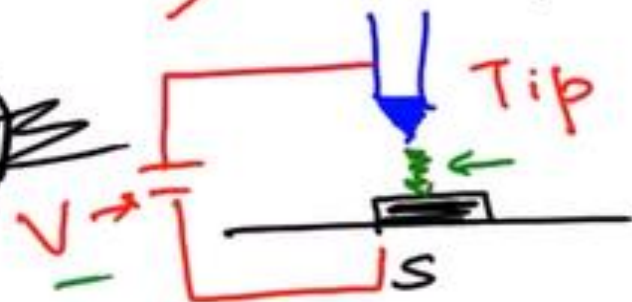
STM

Principle

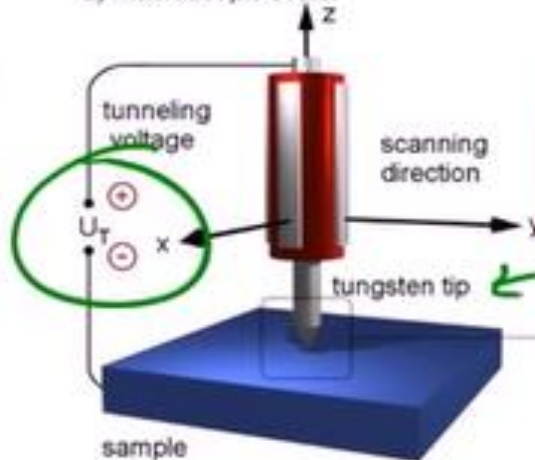
- ① conts. current
- ② const. Height



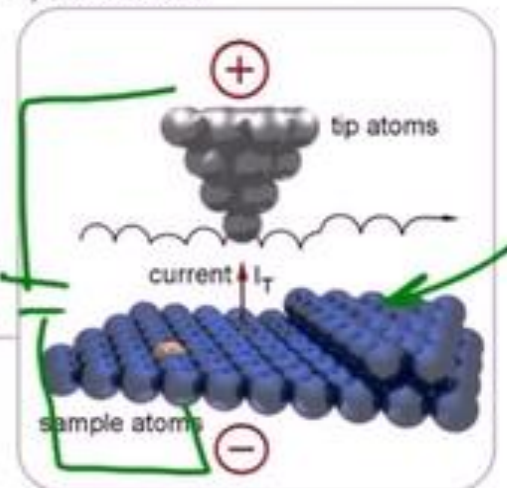
$Q \cdot T$



a) macroscopic scale:

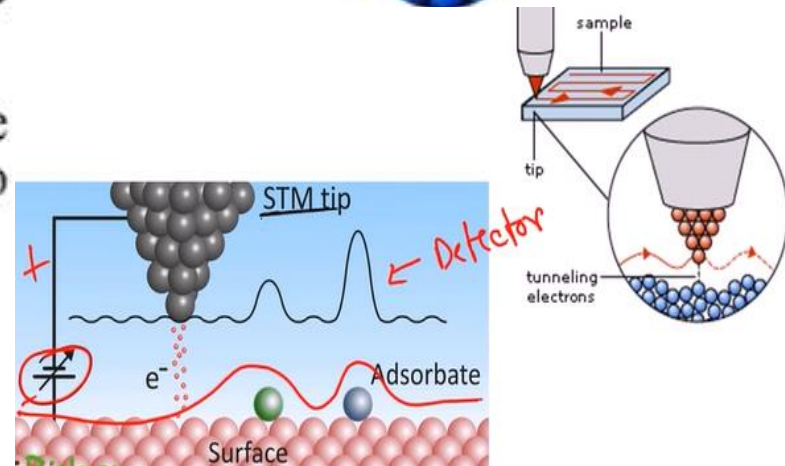
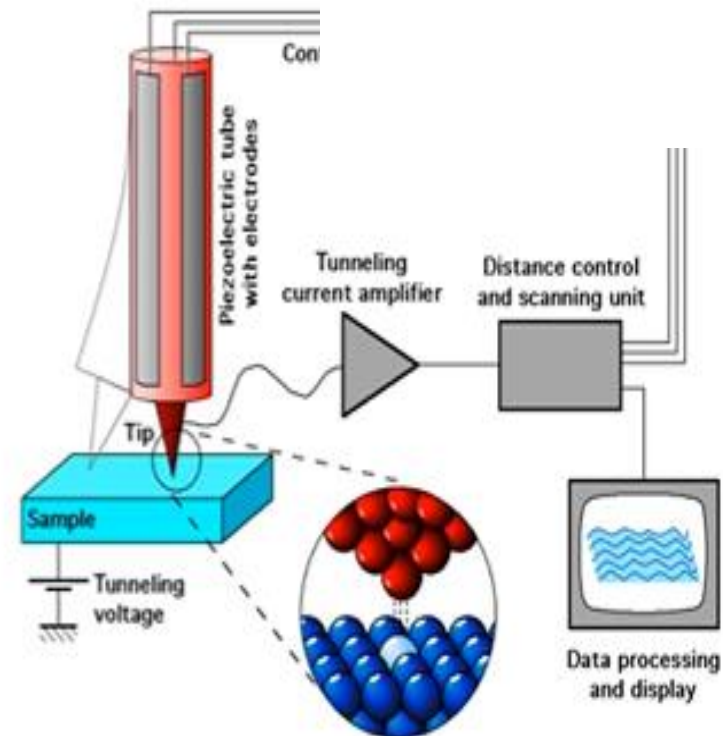


b) atomic scale:

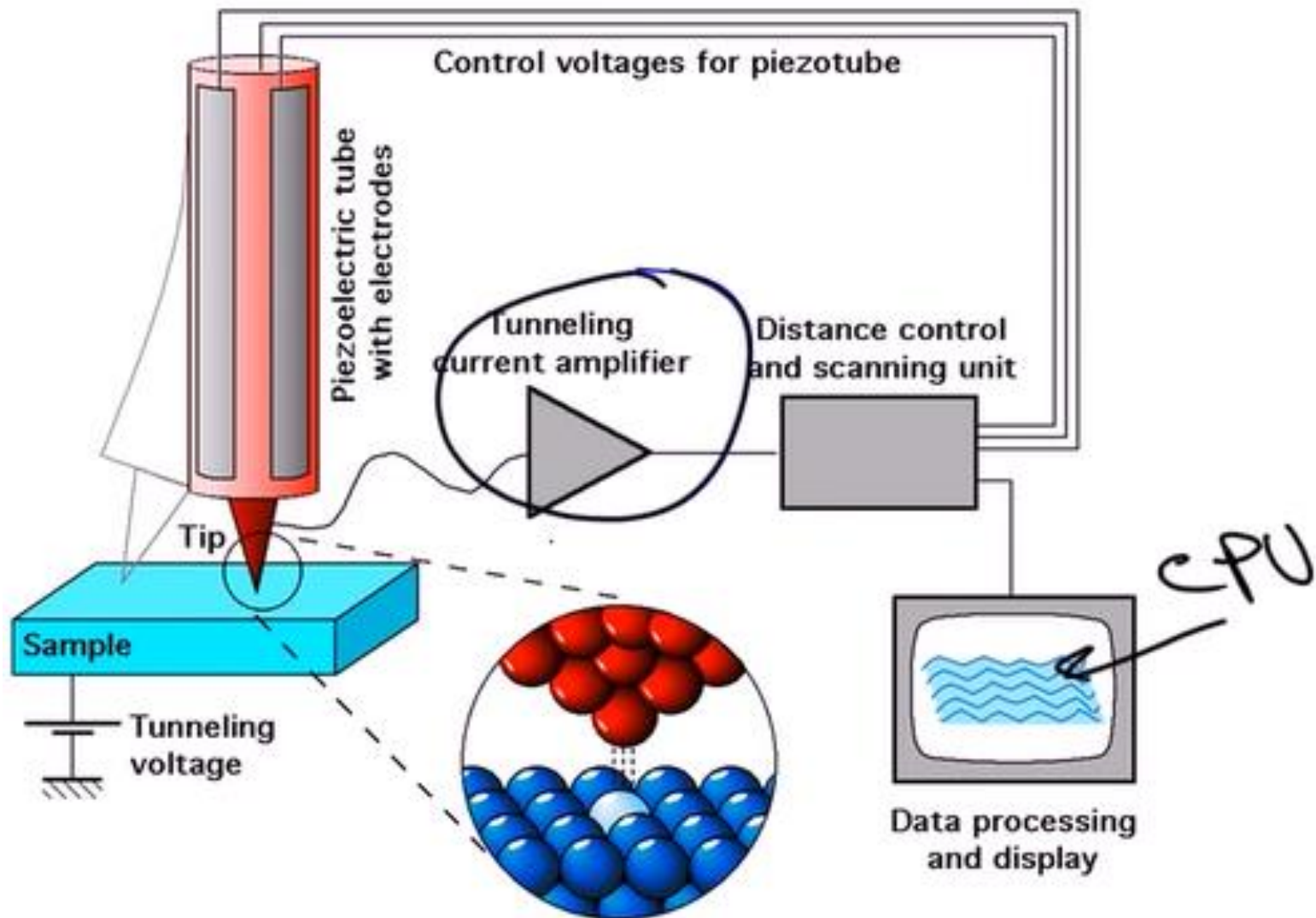


Working

- First, a voltage bias is applied and the tip is brought close to the sample by coarse sample-to-tip control, which is turned off when the tip and sample are sufficiently close. At close range, fine control of the tip in all three dimensions near the sample is typically piezoelectric, maintaining tip sample separation W typically in the $4-7 \text{ \AA}$ ($0.4-0.7 \text{ nm}$) range.
- In this situation, the voltage bias will cause electrons to tunnel between the tip and sample, creating a current that can be measured. Once tunneling is established, the tip's bias and position with respect to the sample can be varied and data are obtained from the resulting changes in current.
- If the tip is moved across the sample in the x-y plane, the changes in surface height and density of states cause changes in current. These changes are mapped in images.
- This change in current with respect to position can be measured itself, or the height, z , of the tip corresponding to a constant current can be measured.



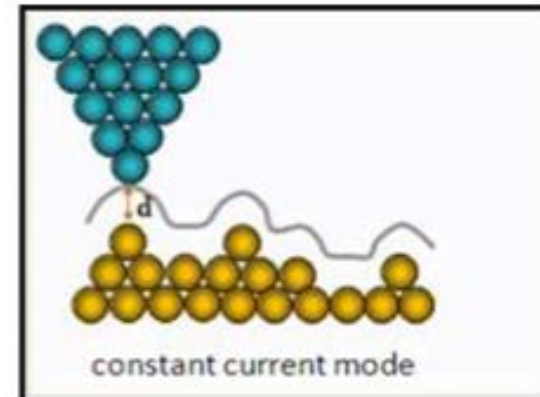
Instrumentation



Modes of operation

➤ Constant Current

- Feedback electronics adjust the height by a voltage to the piezoelectric height control mechanism.
- This leads to a height variation and thus the image comes from the tip topography across the sample and gives a constant charge density surface; this means contrast on the image is due to variations in charge density
- It is a time consuming method compared to the constant height mode as the feed back control has to adjust the current constant according height as the tip moves along the specimen surface.



https://www.phys.sinica.edu.tw/TIGP-NANO/Course/2009_Spring/Class_Notes/CMNS_STM_20090224.pdf

Tunneling Current

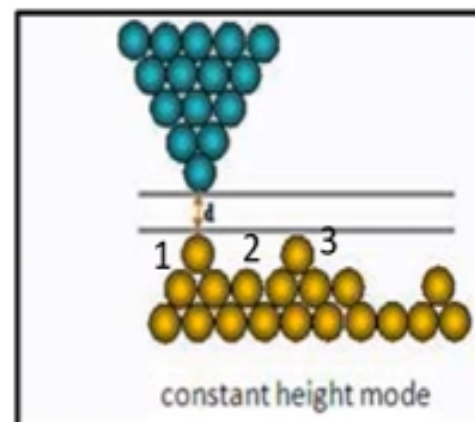


- When a sharp metallic tip is brought very close to the surface of a conductor, an electric current can be detected due to the tunneling of electrons through the air gap.
- In this case, the air gap is considered as the barrier.
- When an electrical voltage V is applied between sample and tip, this tunneling phenomenon results in a net electrical current, the '**tunneling current**'.

Modes of operation

➤ Constant Height

- The voltage and height are both held constant while the current changes to keep the voltage from changing; this leads to an image made of current changes over the surface, which can be related to charge density.
- The benefit to using a constant height mode is that it is faster, as the piezoelectric movements require more time to register the height change in constant current mode.
- However, it is applicable only when sample surface is real flat, the corrugation more than 6 to 7 $\text{\AA}$ will lead tip to crash.
- Generally less preferred due to the risk of damaging the tip.



https://www.phys.sinica.edu.tw/TIGP-NANO/Course/2009_Spring/Class_Notes/CMNS_STM_20090224.pdf

Advantages & Disadvantages

Advantages

- STMs are versatile. They can be used in ultra high vacuum, air, water and other liquids and gasses.
- STMs give three dimensional profile of a surface, which allows researchers to examine a multitude of characteristics, including roughness, surface defects and molecule size.
- Lateral Resolution of 0.1 nm and 0.01 nm of resolution in depth can be achieved.

Disadvantages:

- It is very expensive.
- It need specific training to operate effectively.
- STM need very clean surface, excellent vibration control while operation, single atom tip.